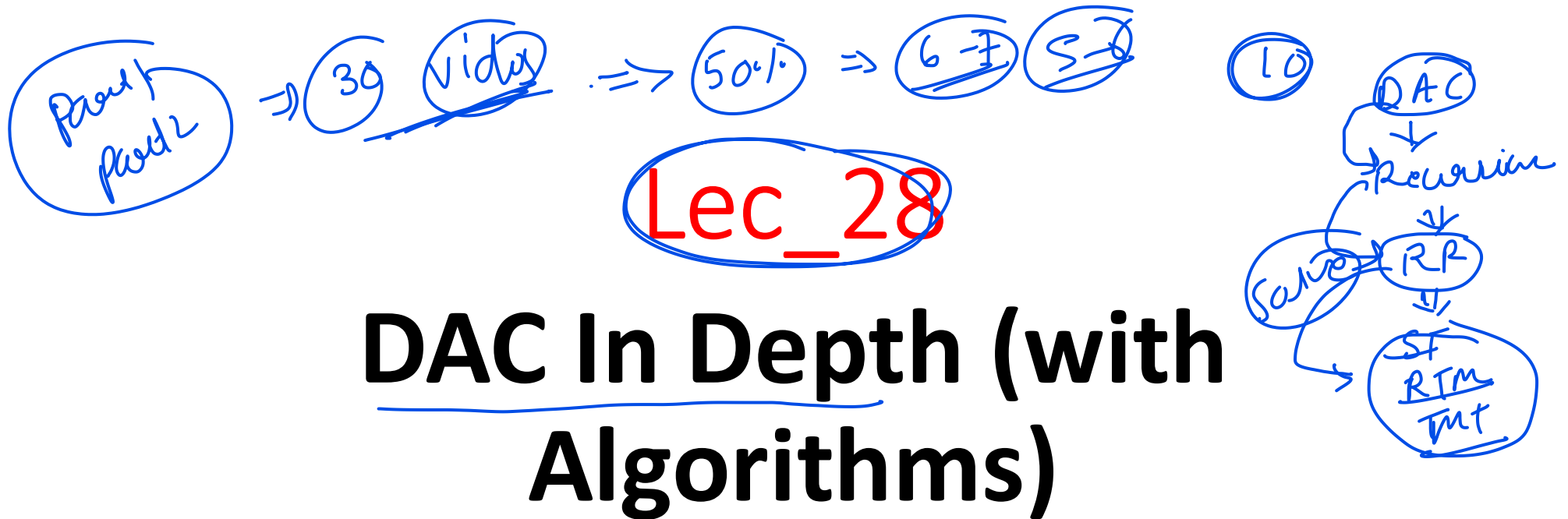




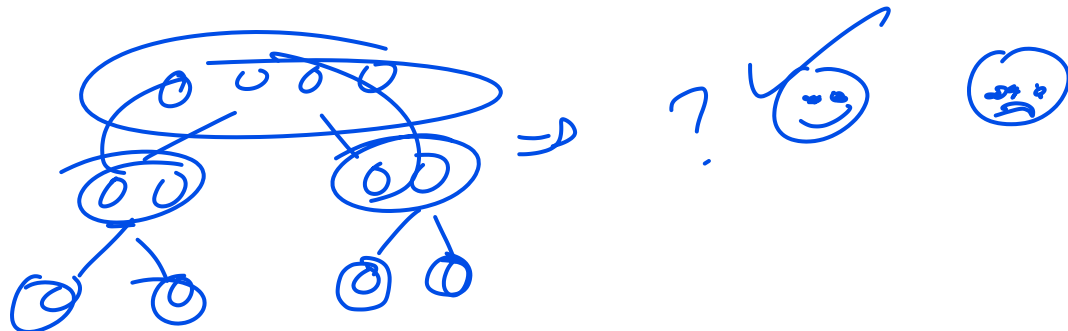
## Basics of DAC: (Divide and Conquer)

✓ Divide the big problem into small problems



• Conquer the Subproblems by Calling Recursively until we will get Sub-problem Solution

• Combine the Subproblem Solutions so that we can get final problem solution



Blueprint

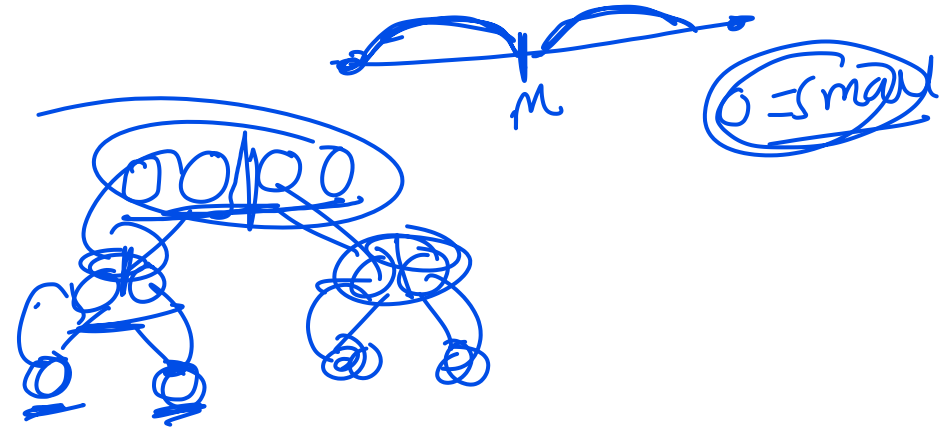
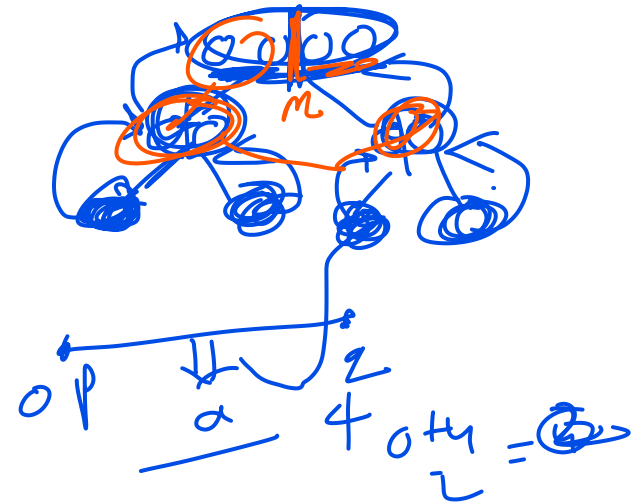
# DAC Abstract Algorithm:

DAC(a, p, q):

a = data structure  
p = starting point  
q = ending point

```

{ if (small(a, p, q))
  { return (solution(a, p, q))
  }
  else
  { m = (p + q) / 2
    s1 = DAC(a, p, m)
    s2 = DAC(a, m + 1, q)
    s = combine(s1, s2)
  }
}
return s;
  
```



$DAK(a, p, z)$

if  $(small(a, p, z))$   
return solution  $(a, p, z)$

else

$m = \frac{p+z}{2}$

$s_1 = DAK(a, p, m)$

$s_2 = DAK(a, m+1, z)$

$s = \text{combination}(s_1, s_2)$   
return  $s$ ;

}



# DAC Time Complexity Analysis:

TC?

Let TC of  $DAC(a, p, z)$  is

$$T(n) = \underbrace{f_1(n)} + 1 + T(n/2) + T(n/2) + \underbrace{f_2(n)}$$

$$T(n) = 2T(n/2) + \underbrace{f(n)} + \underbrace{f(n)+1}$$

$$\boxed{T(n) = 2T(n/2) + f(n)}$$

$$f_1(n) + f_2(n) + 1 = f(n)$$

where

ST, ATM, MT

$DAC(a, p, z)$   
 $\{$   
 $\quad \text{if } (small(a, p, z))$   
 $\quad \quad \text{return solution}(a, p, z)$   
 $\quad \text{else}$   
 $\quad \quad m = \frac{p+z}{2} \Rightarrow c$   
 $\quad \quad \{$   
 $\quad \quad \quad s_1 = DAC(a, p, m)$   
 $\quad \quad \quad s_2 = DAC(a, p, m+1)$   
 $\quad \quad \quad s = combine(s_1, s_2)$   
 $\quad \quad \quad \text{return } s$   
 $\quad \quad \}$   
 $\}$

$f_1(n)$   
 $\{$   
 $\quad small()$   
 $\quad \quad \{$   
 $\quad \quad \quad \text{solution}()$   
 $\quad \quad \quad \{$   
 $\quad \quad \quad \quad \text{contin}()$   
 $\quad \quad \quad \quad \{$   
 $\quad \quad \quad \quad \quad \dots$   
 $\quad \quad \quad \quad \}$   
 $\quad \quad \quad \}$   
 $\quad \}$   
 $\}$

@@@







Join tele

Happy learning

**Thanks for watching**